YEAR 7 · UNIT 4 REVIEW · SCIENCE UNIT 3 · HOMEWORK 10

Faster in January

Part 1 — Mathematics Unit 4 again, with new numbers
Part 2 — Science 3.2, 3.3 and the football model, in words and in decimals

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 30 October 2026

How to do this homework

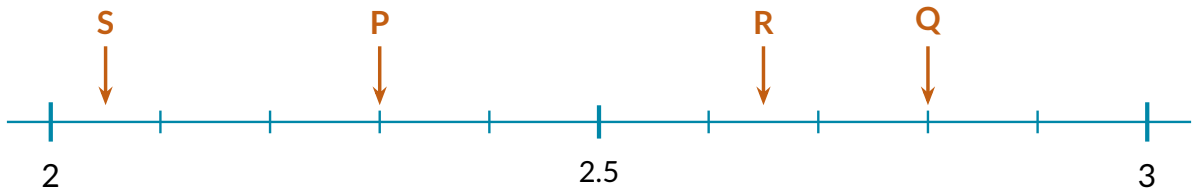
Six sections. A to C are **Unit 4 again**, with new numbers. D to F are this week's science, and **F does it in decimals**.

- **Write the calculation before you write the answer. Estimate first**: the estimate tells you where the point goes.
- **No calculator.** In D, E and F, answer in **full English sentences** where it says so. *"Gravity"* is not a sentence. *"Gravity pulls the planet towards the Sun"* is.

Section A · Unit 4 Review — Comparing, Adding and Subtracting

A1 Read the line

The line goes from **2** to **3**. Every small line is **0.1**. **(a)** Write the number each arrow points to.



P _____	Q _____	R _____	S _____

(b) Write the four numbers in order, **starting with the largest** (the biggest).

_____

A2 Which is bigger?

Write <, > or = in each box. Look at the whole-number parts first. Then **fill with zeros**: 0.5 = 0.50.

(a) 0.5 ☐ 0.45

(b) 2.1 ☐ 2.09

(c) 8.08 ☐ 8.8

(d) 6.4 ☐ 6.400

(e) 15.6 ☐ 9.87

(f) 1.25 ☐ 1.3

A3 Line up the points

Written method

Write the numbers with the **points in a line**. Fill the gaps with **zeros**. The point in the answer goes **straight down**.

(a) 7.8 + 4.56

(b) 23.4 + 0.87 + 5

(c) 16.25 + 3.9

(d) 10 − 3.42

(e) 6.3 − 2.75

(f) 0.9 − 0.36

Section B · Unit 4 Review — Multiplying and Dividing

Where the point goes

× : take the point out and multiply, then **estimate** to put it back. 3.8 × 12: 38 × 12 = 456, and 4 × 12 = 48, so **45.6**.
÷ : short division. The point goes **straight up**. If a digit is too small to divide, write **0**. To divide by a 2-digit number, write its **times table** first.

B1 Point out, point back

Fill in the table. The first one is done for you.

Calculation	Estimate	Without the point	Answer
2.4 × 13	2 × 13 = 26	24 × 13 = 312	31.2
0.6 × 9			
3.45 × 6			
7.2 × 15			
2.8 × 23			

B2 The point goes straight up

Use short division. For **(e)** and **(f)**, write the times table first.

(a) 9.45 ÷ 5

(b) 4.32 ÷ 4

(c) 0.72 ÷ 9

(d) 7.3 ÷ 2

(e) 52.5 ÷ 15

(f) 81.6 ÷ 24

Section C · Unit 4 Review — Words, Quick Ways and Money

C1 From English to a calculation

Word help — the words that choose the operation

each	for one of them. *3 pens at 2 dollars each*: 3 × 2
altogether	all of it, the total
twice as far	2 times as far
what must you add to 4 to make 7?	7 − 4. It says *add*, but you **subtract**
3 less than 10	10 − 3. The **second** number comes **first**

Write the **calculation first**, then the answer. Give units.

(a) What must you add to 4.65 to make 7?

Calculation: _____ **Answer:** _____

(b) A ribbon 2.4 m long is cut into 16 equal pieces. How long is each piece?

Calculation: _____ **Answer:** _____

(c) Mr Bowen walks 2.5 km on Monday. On Tuesday he walks twice as far. How far does he walk altogether?

Calculation: _____ **Answer:** _____

(d) What is 3.5 less than 10?

Calculation: _____ **Answer:** _____

C2 Use a quick way

Quick ways

× 0.5 = **half of** × 0.25 = **a quarter of** × 0.07 = ÷ 100, then × 7
× 9: × 10, then **take away** one lot. × 11: × 10, then **add** one lot.
Partitioning: split the number. 4.3 × 6 = 24 + 1.8

Write the quick way, then the answer.

(a) 4.7 × 9 = _____ = _____

(b) 6.2 × 11 = _____ = _____

(c) 5.3 × 4 = _____ = _____

(d) 64 × 0.25 = _____ = _____

(e) 300 × 0.07 = _____ = _____

(f) 9.6 × 0.5 = _____ = _____

C3 Change from 10 dollars

Money has **2 decimal places**: 7.8 dollars is written 7.80 dollars.

Mr Bowen buys 3 bánh mì at 1.35 dollars each and 2 iced coffees at 1.90 dollars each. He pays with a 10-dollar note. How much change does he get?

C4 Mr Bowen's homework

Only **one** of these is right. Find it, and correct the other three.

(a) 4.5 + 2.7 = 6.12 _____

(b) 9.6 ÷ 3 = 3.2 _____

(c) 3.40 > 3.4 _____

(d) 60 × 0.4 = 2.4 _____

C5 Now you write the question

The rules

A **real** situation, in full sentences, ending in a **question**. Use a **word** from C1, give **units**, and come out at the answer given.

Write a word problem for $12 \times 2.40 = 28.80$

Section D · Science 3.2 — How the Solar System Formed

D1 Key words

Write the letter of the right meaning next to each word.

nebula		**A**	an idea that can be tested
hypothesis		**B**	one flat surface
evidence		**C**	a cloud of dust and gas in space
model		**D**	facts from observations or experiments
limitation		**E**	something a model cannot do, gets wrong or leaves out
plane		**F**	a way of showing something that is hard to observe directly

D2 Support or contradict?

The **hypothesis**: **The Sun and the planets formed from one spinning cloud of dust and gas.**

Does each fact **support** the hypothesis, or **contradict** it? Tick one box.	**supports**	**contradicts**
All the planets orbit the Sun in the same direction.		
All the orbits are in the same plane.		
Venus spins the opposite way to most planets.		
HL Tauri is a young star with a flat disc of dust round it.		
Uranus spins on its side.		
Most moons orbit their planet in the same direction.		

(b) Two facts contradict the hypothesis. So is it wrong? Answer in a full sentence. Use the words **most of the evidence**.

D3 From a cloud to the Sun and planets

(a) Put the four steps in order. Write **1**, **2**, **3** and **4** in the boxes.

☐ The hot middle becomes the Sun.

☐ Gravity pulls the nebula together. It spins faster.

☐ Dust sticks together and grows into planets.

☐ The nebula flattens into a spinning disc.

(b) Why do all the planets orbit the Sun in the same direction? Answer in a full sentence. Start: **"They all formed from …"**

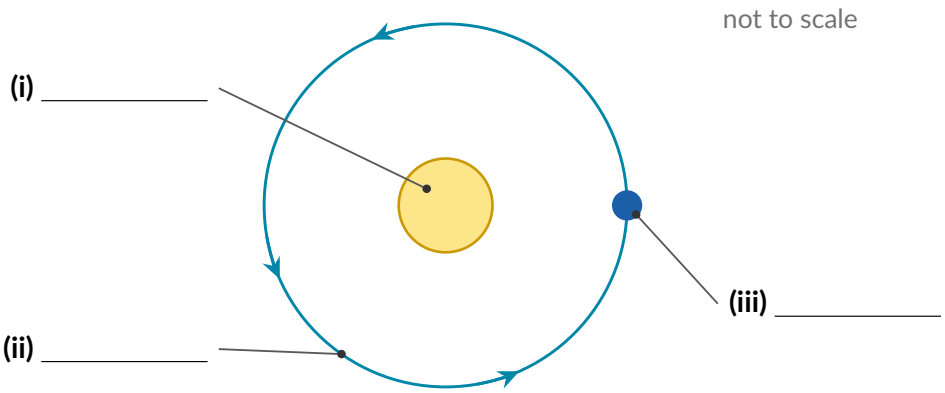
(c) Mercury, Venus, the Earth and Mars are small and rocky. Why? Start: **"Near the Sun, it was too hot for …"**

(d) The Sun has **99.8%** of all the mass in the Solar System. % means *out of 100*. How many % are left for everything else? _____

Section E · Science 3.3 — Movement in Space

E1 What keeps a planet in orbit?

(a) Label the drawing. Use these words: **Sun** · **orbit** · **planet**



(b) Draw an **arrow** on the planet to show the pull of the Sun's **gravity**.

(c) The arrows show which way the planet moves. Now the Sun's gravity **switches off**. Draw a **dashed line** to show where the planet goes.

E2 Which forces?

Which forces act on each one? Tick **every** box that is true.	**gravity**	**air resistance**
The Earth, orbiting the Sun		
The Moon, orbiting the Earth		
A ball falling from the balcony		
A plane landing on a runway		

(b) Why do the planets never slow down? Start: **"Space is a …, so …"**

Section F · Science in Decimals — Faster Near the Sun

Speed, from Science 3.3

Speed: how fast something moves. **km/s** = kilometres **per** second.
As the distance from the Sun **increases**, the speed of the planets **decreases**. The Sun's pull is **weaker** far away.

F1 How much faster? How many times as fast?

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
speed (km/s)	**48**	**35**	**30**	**24**	**13**	**10**	**7**	**5**

Word help — two questions that look the same

how much faster? the difference. **Subtract**

how many times as fast? **divide** the bigger speed by the smaller one

Fill in the table. The first row is done for you.	**How much faster?**	**How many times as fast?**
Mercury and Neptune	48 − 5 = 43 km/s	48 ÷ 5 = 9.6
Mercury and the Earth		
Mars and Saturn		
The Earth and Mars		

F2 The Sun is a football

In the lesson the Sun became a football, and everything else shrank by the same amount.

	the Moon	**the Earth**	**Neptune**	**Jupiter**	**the Sun**
Model width	0.5 mm	2 mm	7.8 mm	2.3 cm	22 cm
Like	a grain of sand	a pinhead	a pea	a grape	a football

(a) Write the five model widths in order of size, starting with the **smallest**. Remember: **1 cm = 10 mm**.

(b) In the model, the Earth is **24 m** from the football. Mr Bowen builds the model in his classroom, which is **8.5 m** long. He puts the football against one wall. How far past the other wall must the Earth go?

Calculation: _____ **Answer:** _____

(c) Give **one limitation** of the football model. Start: **"One limitation of the football model is …"**

F3 Faster in January

The Earth's orbit is *almost* circular. In January the Earth is a little **closer** to the Sun than in July.

	Distance from the Sun	Speed
January	147.1 million km	30.3 km/s
July	152.1 million km	29.3 km/s

(a) How much closer to the Sun is the Earth in January?

Calculation: _____ **Answer:** _____

(b) In which month is the Earth faster? Why? Use the words **closer** and **pull**. Start: **"The Earth is faster in …, because …"**

(c) How much faster is the Earth in January than in July?

Calculation: _____ **Answer:** _____

(d) Find the **mean** of the two speeds.

Calculation: _____ **Answer:** _____

(e) Round your mean to the nearest whole number. _____

In the lesson, the Earth's speed was **30 km/s**. What do you notice?

Answer Key — Teacher Copy

A1 Read the line

(a) P 2.3 · Q 2.8 · R 2.65 · S 2.05
(b) 2.8, 2.65, 2.3, 2.05

R and S sit halfway between two lines; HW 9's A1 asked the class to *place* 5.45 and 5.05, and this asks them to *read* them. Expect 2.5 for S from a student who sees "halfway" and writes a 5 in the tenths, and 2.6 or 2.7 for R. (b) runs from the *largest* on purpose — HW 9 always ran from the smallest. A list that starts with 2.05 is right in the wrong direction: ask what *largest* means.

A2 Which is bigger?

(a) > (b) > (c) < (d) = (e) > (f) <

(a) and (f) are the same trap from two sides: the number with more digits is the smaller one (0.50 > 0.45, 1.25 < 1.30). (c) is "eight point zero eight" against "eight point eight"; anyone who reads both as "eight point eight" has dropped the zero. (e) is the whole-number part first: 15 > 9, and the digits after the point do not matter.

A3 Line up the points

(a) 12.36 (b) 29.27 (c) 20.15 (d) 6.58 (e) 3.55 (f) 0.54

(a) and (c) are the alignment check: lining up the right-hand digits gives 7.8 + 4.56 = 5.34 and 16.25 + 3.9 = 16.64. (b) has a whole number with no point: 5 = 5.00. (d), (e) and (f) need the zeros *written* first: 10.00, 6.30, 0.90. Expect 7.42 for (d) from a student who took 10 − 3 and copied the .42.

B1 Point out, point back

0.6×9	$1 \times 9 = 9$	$6 \times 9 = 54$	**5.4**
3.45×6	$3 \times 6 = 18$	$345 \times 6 = 2070$	**20.7**
7.2×15	$7 \times 15 = 105$	$72 \times 15 = 1080$	**108**
2.8×23	$3 \times 20 = 60$	$28 \times 23 = 644$	**64.4**

Rows 2 and 3 end in a zero once the point is back (20.70, 108.0); accept them with or without it. Row 3 is the one to watch: $72 \times 15 = 1080$, and a student who puts back a place for *each* number writes 10.80. Only 7.2 has a decimal place. The estimate, 105, settles it.
Row 1 is 0.6×9, not 6×9: expect 0.54 from anyone counting all the digits instead of the decimal places.
Any sensible estimate is right ($2.8 \times 23 \approx 3 \times 23 = 69$ is fine).

B2 The point goes straight up

(a) 1.89 (b) **1.08** (c) **0.08** (d) **3.65** (e) 3.5 (f) 3.4

(b) needs one 0 (4 into 3 will not go); expect 1.8. (c) needs two: 9 into 0, then 9 into 7, will not go. Expect 0.8, which is $0.72 \div 0.9$.
(d) runs out of digits: write 7.30 and keep going. "3.6 remainder 1" is the usual stop.
(e) and (f): the table is the method. $15 \times 3 = 45$, 7.5 left = 15×0.5. $24 \times 3 = 72$, 9.6 left = 24×0.4. A student with the right answer and no table probably knew it; one with a wrong answer and no table needs telling to write the table.

C1 From English to a calculation

(a) 7 − 4.65 = 2.35 (b) 2.4 ÷ 16 = 0.15 m
(c) 2.5 × 2 = 5, 2.5 + 5 = **7.5** km (d) 10 − 3.5 = 6.5

Mark the calculation, not the answer. This is the English question.
(a) says *add* and is a subtraction. 4.65 + 7 = 11.65 is the English error, not an arithmetic one: have the student read the word-help line aloud.
(d) is "3.5 less than 10": 3.5 − 10 is reading left to right.
(c) is two steps and *altogether* is the second. Expect 5 km, which is Tuesday alone.
(b) is the 16 table: 16 × 0.1 = 1.6, 0.8 left = 16 × 0.05. Expect 1.5 from a student who did 24 ÷ 16 and put the point back in the wrong place; the estimate (2.4 m in 16 pieces is much less than 1 m each) catches it.

C2 Use a quick way

(a) 47 − 4.7 = 42.3 (b) 62 + 6.2 = 68.2 (c) 20 + 1.2 = 21.2
(d) a quarter of 64 = 16 (e) 3 × 7 = 21 (f) half of 9.6 = 4.8

(a) needs 47.0 − 4.7. Expect 43.7 from a student who took 47 − 4 and copied the .7 across — the A3(d) error again.
(e) divides by **100**, not 10. Expect 210 (30 × 7).
Accept a correct written method for any of them, but the column is "the quick way": ask for the quick way.

C3 Change from 10 dollars

3 × 1.35 = 4.05, 2 × 1.90 = 3.80, 4.05 + 3.80 = 7.85, 10.00 − 7.85 = **2.15** dollars

Four steps. The usual slip is to stop at 7.85 and call it the change; ask what the question asked for. 3.8 for the coffees is right but is money, so 3.80.
10.00 − 7.85 needs the zeros written. Expect 3.85 from a student who took 10 − 7 = 3 and copied the .85 — A3(d) and C2(a) again.

C4 Mr Bowen's homework

(a) **wrong** — 7.2. He added 4 + 2 and 5 + 7 separately; the 12 tenths carry.
(b) **right.**
(c) **wrong** — 3.40 = 3.4. The zero at the end does not change anything.
(d) **wrong** — 24. × 0.4 is ÷ 10, then × 4: 6 × 4.

One trap each from 4.2, 4.4, 4.1 and 4.5. Whichever one a student gets wrong, send them back to A3, B2, A2 or C2 rather than explaining again.
(b) is right on purpose and is the one with most room for doubt: a student who expects every division to need a 0 will "correct" it to 3.02.

C5 Now you write the question

Check, in order: a real situation; full sentences; ends in a question mark; uses a word from C1; gives units; and it really comes out at 28.80.

A good answer: "Mr Bowen buys 12 notebooks at 2.40 dollars each. How much does he pay altogether?"
The common failure is a question whose answer is 0.20: "12 friends share 2.40 dollars". Read it back to them and ask what the calculation in the box does.
Units need not be dollars — "12 bags of rice, 2.40 kg each" is just as good — but 28.80 with two decimal places is a strong hint.

D1 Key words

nebula **C** · hypothesis **A** · evidence **D** · model **F** · limitation **E** · plane **B**

These are the 3.2 and Models write notes, word for word. Hypothesis and evidence (A and D) are the pair most often swapped; *plane* is the one most often left blank ("not an aeroplane").

D2 Support or contradict?

Supports: rows 1, 2, 4, 6. Contradicts: rows 3 (Venus), 5 (Uranus).
(b) "No. Most of the evidence supports the hypothesis, so it is not wrong. It is just not proven."

(b) is the deck's Question 4 ("most, but not all, of the evidence supports it"). Accept any sentence that says the hypothesis is kept because most evidence supports it. A student who says "yes, it is wrong" has heard *contradict* as *prove wrong*; the cake on deck slide 6 is the reminder — two clues still supported the cat.
Row 4 (HL Tauri) is the one students hesitate over. It supports: a young star with a flat disc is the hypothesis happening today.

D3 From a cloud to the Sun and planets

(a) In the order printed: **3, 1, 4, 2**.
(b) "They all formed from one spinning cloud, so they all still go round the same way."
(c) "Near the Sun, it was too hot for ice, so only rock and metal were left to make planets."
(d) 100 − 99.8 = **0.2**%

(a) The printed order is shuffled; read it as "the hot middle is 3". The usual error is putting the Sun first, which is the order of importance, not of time.
(c) is the Models deck's "Rocky or giant?" slide; "too hot for ice" is the point. Accept "too hot for gas" only if they also say the planets are made of rock and metal.
(d) needs 100.0 − 99.8. Expect 1.8 (100 − 99, then the .8 copied) or 1.2 from anyone who did not write the zero. Every planet, moon, asteroid and comet together is two tenths of one percent: that is worth saying aloud.

E1 What keeps a planet in orbit?

(a) (i) Sun · (ii) orbit · (iii) planet
(b) An arrow from the planet pointing **to the left**, at the Sun.
(c) A straight dashed line going **straight up the page** from the planet — along the direction it was moving, touching the orbit, not crossing it.

(c) is the class vote from the lesson (left hand "away from the centre", right hand "straight on"), and the wrong answer is the same here: a line straight out to the right, away from the Sun. It means the student thinks something was pushing the planet outwards. Nothing was — gravity was pulling it in, and when gravity stops the planet simply keeps going the way it was going.
A curved dashed line that keeps going round is the other error: without gravity nothing turns it.

E2 Which forces?

Earth: gravity · Moon: gravity · falling ball: gravity *and* air resistance · plane: gravity *and* air resistance
(b) "Space is a vacuum, so there is no air resistance to slow the planets down."

Row 1 is the deck's Question 6, and row 2 is the same answer for the Moon: gravity only. Expect air resistance ticked for the Earth by students who picture the Earth's own air; it is the planet moving through space that matters, and space has no air.
Rows 3 and 4 are on Earth, where there is air. The plane is the photo from the air resistance slide.
(b): "vacuum" and "no air resistance" are the two things to look for.

F1 How much faster? How many times as fast?

Mercury and the Earth	48 − 30 = 18 km/s	48 ÷ 30 = 1.6
Mars and Saturn	24 − 10 = 14 km/s	24 ÷ 10 = 2.4
The Earth and Mars	30 − 24 = 6 km/s	30 ÷ 24 = 1.25

The two columns are the point. A student who writes the same kind of answer in both has not read the questions. "How many times" has no unit: 1.6, not 1.6 km/s.
30 ÷ 24 is the review's 2-digit division: 24 × 1 = 24, 6 left, 60 ÷ 24 = 2 remainder 12, 120 ÷ 24 = 5, so 1.25. Expect 1.2 remainder 12.
48 ÷ 30 and 24 ÷ 10 need a 0 written: 48.0, 24.0. Expect 1.6 to come out as "1 remainder 18".

F2 The Sun is a football

(a) 0.5 mm, 2 mm, 7.8 mm, 2.3 cm, 22 cm (b) 24 − 8.5 = **15.5** m
(c) Any one: nothing moves (the planets do not orbit or spin); the football is not hot and gives no light; most of the planets and moons are missing; a 0.5 mm Moon is too small to see.

(a) is a units trap. A student who ignores the units puts 2.3 before 7.8 — Jupiter smaller than Neptune — and the grape and the pea in the table say otherwise. 2.3 cm is 23 mm.
(b) needs 24.0 − 8.5. The pinhead goes 15.5 m out into the corridor, which is the joke and also the point of the Models lesson: the Solar System is mostly empty.
(c) is the Cambridge question. Any honest limitation in a full sentence is right; "it is not real" is not a limitation — no model is real.

F3 Faster in January

(a) 152.1 − 147.1 = **5** million km
(b) "The Earth is faster in January, because it is closer to the Sun and the Sun's pull is stronger."
(c) 30.3 − 29.3 = **1** km/s (d) 30.3 + 29.3 = 59.6, 59.6 ÷ 2 = **29.8** km/s
(e) **30**. "It is the same as the Earth's speed in the lesson."

This is the question the packet is built around. It is 3.3's exit question with real numbers (NASA: 147.1 and 152.1 million km; 30.29 and 29.29 km/s). The rule from 3.3 is that closer planets move faster, and here it holds for one planet over one year.
(e) is the reveal: the book's 30 km/s is the Earth's *mean* speed, rounded. Worth putting on the board: the table in F1 is a table of means. Any sentence that says the two are the same earns (e); one that says *why* ("the lesson used the mean") is the best answer on the packet.
(a) and (c) both come out as whole numbers written as decimals (5.0, 1.0); accept either. (b) is 3.3's "closer planets move faster" turned on one planet: accept any sentence that says January and gives the reason. "Because it is closer" alone is half of it — ask *why* closer means faster, and look for *pull*.
Do not let this become a lesson on seasons. January is not the hottest month, and the reason (the tilt of the Earth) has not been taught. If someone asks, that is a good question to leave open.